

MyDeliveryServiceImpl Module

Provides concrete implementations of service APIs, models, integration ACLs, helpers, HTTP utilities, reporting assembly, and AspectJ concerns.

Packages

- `model.impl` : POJO implementations of API model interfaces.
- `service` : Core service façade implementations.
- `service.acl.*` : Anti-Corruption Layer adapters for external systems (APC, OSC, StreamServe, common codes).
- `service.aspectj` : Cross-cutting concerns (caching aspect).
- `service.helper` : Utility builders and validation helpers specific to service logic.
- `service.http` : HTTP client configuration & execution utilities.
- `service.integration` : Domain integration aggregates & mutable helper objects for redelivery processes.
- `service.reports` : Report data assembly and printing support.
- `service.acl.streamserve.request` : JAXB-mapped XML payload/request/response classes.

Model Implementations

- `DefaultAvailableOptionsModel` : Holds computed redelivery options & potential validation violations.
- `DefaultBaseResponseModel` : Base response implementation (implements `BaseResponseModel` plus common serialization helpers).
- `DefaultConfirmationModel` : Concrete confirmation details (reference IDs, timestamps, status codes).
- `DefaultCountryMetadataModel` : Country rules (postcode pattern, phone prefix, feature toggles).
- `DefaultDatesModel` : Backing for selectable dates (internally ensures ordering, uniqueness, earliest date retrieval).
- `DefaultDepotAddressModel` : Concrete depot address fields (name, street, city, postal code, country).
- `DefaultDepotDateAndTimesModel` : Opening/collection window representation for a depot date.
- `DefaultPrintConfirmationModel` : Data & formatted date strings for confirmation printing.
- `DefaultPrintPTLModel` : Data & formatted date/barcode strings for PTL form printing.
- `DefaultSelfCollectionModel` : Aggregates depot list, selection state and available collection dates.
- `DefaultViolationsModel` : Maintains a list of `MyDeliveryRuleViolation` grouped for responses.

Core Service

- `DefaultMyDeliveryServiceBase` : Abstract/base class consolidating common utilities, property loading, locale resolution, and error mapping logic shared by services.
- `DefaultMyDeliveryService` : Concrete implementation of `MyDeliveryService` ; orchestrates validation, option determination, consignment retrieval, redelivery confirmation, printing request building, and integration calls to ACLs.

Helpers

- `LocaleHelper` : Resolves locale / language settings via system parameters & request hints.
- `PtlFormDataCreator` : Assembles `PtlFormData` from consignment + user input (dates, charges, references) with formatting.
- `ValidationHelper` : Wraps validation routines, collecting `ErrorContainer` entries into `ViolationsModel` results.

AspectJ

- `Caching` : Aspect providing method-level caching (likely around read-only service methods) with logging and key generation.

HTTP Utilities

- `HttpConfig` : Builds and configures Apache HttpClient (timeouts, credentials, proxy, pooling).
- `HttpProcessor` : Executes HTTP calls with retry / timeout handling, status code validation, entity reading, and logging.

ACL (Anti-Corruption Layer) Adapters

- `apc.DefaultAPCAcl` : Wraps APC system calls (e.g., address/consignment lookups) adding validation & logging.
- `commoncodes.DefaultCommonCodesAcl` : Fetches and caches reference/common code data (e.g., country lists, option flags).
- `osc.DefaultOSCAcl` : Calls OSC service for operational service capabilities / country info; implements `OSCAcl` interface.
- `streamserve.DefaultStreamServeAcl` : Sends document generation requests (labels, confirmations) to StreamServe, handling marshalling and response parsing.

StreamServe Support

- `PayloadCreator` : Builds XML request payload objects (`XmlRequest` , child elements) from domain data.
- `RequestHandler` : Marshals JAXB objects to XML bytes & logs payload.
- `ResponseHandler` : Interprets HTTP responses, captures errors, and extracts violation details.

StreamServe JAXB Request/Response Classes

- `request.Address` , `Sender` , `Receiver` : Parties involved in document generation (name, address lines, contact fields).
- `request.Consignment` : Shipment identification (numbers, piece counts, weight).
- `request.Option` : Service option element inside product.
- `request.Product` : Product/service details with option list wrapper.
- `request.XmlRequest` : Root request envelope containing consignments, parties, metadata.
- `request.XmlResponse` : Root response capturing document generation outcome/status.

Integration Domain / Redelivery Workflow

- `DefaultAddress` : Mutable address implementation with validation utilities; used in construction of requests.
- `DefaultConsignmentDetails` : Implementation linking underlying domain consignment to API `ConsignmentDetails` .
- `DefaultDeliverySystemParameter` : Simple key/value / typed parameter representation.
- `DefaultDepotAvailability` : Availability windows for self collection depots (dates & capacity info).
- `DefaultMyDeliveryPhoneNumber` : Implementation including formatting / sanitization logic.
- `DefaultPtlFormData` : Concrete `PtlFormData` including numeric fields (charges) and references.
- `DefaultRedeliveryConsignment` : Aggregate containing consignment base data, selected redelivery path, allowed options, and validation state.
- `DefaultRedeliveryConsignmentHelper` : Utility to derive permissible options based on consignment status, service codes, country rules.
- `DefaultRedeliveryConsignmentParameters` : Carries user selections (date, address variant, instructions) before constructing a request.
- `DefaultRedeliveryRequest` : Immutable request built from bean/parameters ready for confirmation.
- `DefaultRedeliveryRequestBean` : Mutable user form bean with raw input subject to validation.
- `DefaultRedeliveryRequestFactory` : Creates `RedeliveryRequest` from `RedeliveryRequestBean` after applying business rules.
- `DefaultSelfCollectionDepotInformation` : Collection of depots available for self collection (with geocodes & distances).
- `DeliveryRequestEnquiryService` : Interface to determine redelivery options for a consignment id.
- `MutableRedeliveryConsignment` : Internal mutable variant of `RedeliveryConsignment` for assembly.
- `MutableRedeliveryRequest` : Internal mutable representation used during factory construction.

Reporting

- `MyDeliveryConfirmationDetail` : Value object aggregated for JasperReport (label fields, localized strings, images, map of dynamic values).
- `MyDeliveryConfirmationDetailBuilder` : Builds detail objects from `RedeliveryRequestBean` & other model data.
- `MyDeliveryConfirmationDetailLabels` : Resolves localized label names via CMS / message source.
- `MyDeliveryReportPrinter` : Implements `ReportPrinter` ; compiles/fills JasperReports with provided detail/label data.

Cross-Cutting Notes

- `ReflectionToStringBuilder` usage across models ensures consistent logging string form.
- `Net.sf.oval` annotations provide declarative validation in model & request classes.
- `SimpleDateFormat` usage suggests date formatting not yet migrated to thread-safe `DateTimeFormatter`; caution for concurrency.

Extension Points

- Add new external system: create new `service.acl.<system>` package with interface in API module & implementation here.
- Introduce caching for a method: annotate or update `Caching` aspect pointcut expressions.
- Add new report: create builder, labels provider updates, and add printer method in `MyDeliveryReportPrinter` .